Natalie E. **Hagan**

Software Engineer · PhD Candidate · Cyber Security

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Education

Dakota State University — *Madison, SD*

Ph.D. Cyber Operations — *May 2028 (est)*

- Dissertation: *Architecture Neutral Vulnerability Identification in Binaries Using Natural Language Processing Techniques*

Johns Hopkins University — *Baltimore, MD*

M.S. Electrical Engineering — *May 2004*

University of Maryland — *College Park, MD*

B.S. Electrical Engineering — *May 2001*

- Magna Cum Laude
- Eta Kappa Nu, Tau Beta Pi honor societies

Skills and Tools

Languages	C, C++, C#, Assembly (x86, PPC, and others), Python, MATLAB
Software	Ghidra, WinDbg, IDA Pro, Wireshark/Ethereal, SVN, HG
Hardware	Electrical Engineering laboratory instruments including Signal Generators, Vector Analyzers, Oscilloscopes, SDRs
Operating Systems	Linux, Windows, OS X

Work Experience

Nexxis Solutions — *Gambrills, MD*

Senior Cyber Security Capability Engineer & CNO Lead — *Sep. 2021 - Present*

- Capability Developer, Delivery Order Lead
  - Product Architect / Product Owner for Section's top priority effort:
    - Defined and implemented the overall architecture and technical strategy for 6 government-contractor blended teams across multiple states to improve release rate for CNO tools and techniques.
    - Collaborated with team leads to ensure alignment with mission goals, scalability needs, and non-attribution of capabilities.
    - Led cross-team initiatives for enhancing capabilities, identifying capability gaps, and standardizing systems engineering processes.
    - Key stakeholder in strategic planning and road map development by providing staffing placement, technical development input, identifying blockers, and directing mitigations and courses of action.
    - Provided technical evaluation and direction in the design and implementation of CNO tools and techniques across teams.
    - Partner with IC and Second Party partners in the development and use of CNO capabilities.
    - Focused the assessment, triage, and development of remaining tasks for highly anticipated releases of a complex capabilities, resulting in the release of an inaugural tool followed rapidly with 3 additional releases within 3 months.
  - Technical Lead for Delivery Order:
    - Technical lead of delivery order creating offensive cyber capabilities against low-level, high availability network technology targets.
    - Team consisted of 40 multi-disciplinary personnel across 4 sites in 3 states.
    - Responsibe for reporting technical status, contract deliverables, and aligning contractor personnel to mission needs.
    - Increased staffing percentage from 70% to 100% staffed through working with the prime contractor to recruit, interview, and select the appropriate staff for technical needs.
  - CNE capability developer:
    - Performed Reverse Engineering through static and dynamic analysis to understand functionality in a high-availability kernel-level embedded system with limited documentation.
    - Created tailored operational capabilities in C and Assembly, serving as the primary developer for three major features to enable access to an important set of targets.
    - Authored and maintained documentation required for design reviews, release, and operational use of the product, leading meetings with operations and technical leadership to obtain approvals.
    - Provided technical advice, expertise, and support to Operations personnel.
  - Primary technical author for TTO in $1.2B re-competition proposal, resulting in the award of the contract.

Johns Hopkins Applied Physics Lab — *Laurel, MD*

Principal Cyber Capability Engineer & Assistant Group Supervisor — *Jan. 2007 - Aug. 2021*

- **Technical**
  - CNE capability SME embedded with US CYBERCOMMAND providing technical direction and leadership for $1B congressional funding targeting a major operational challenge.
  - Researched and developed CNE techniques using RE and non-traditional mechanisms to reduce risk of PSP identification.

- Created cohesive proof of concept capability using multiple APL-owned capabilities in support of an inter-agency collaboration effort to improve communications and demonstrate potential CONOPs linking capabilities across the IC and DoD.
- Principal developer for an operational tool that covertly modified the operation of commodity hardware including significant RE of the HW.
- SME on special purpose receiver, implementing and honing a tracking algorithm in C# to reduce processing time by 95% and reduce false alarm rate.
- Researched and ported two G975 advanced forward error correction standards to Matlab for evaluation of current communications systems on a classified task.
- Led preparation for two OCONUS field tests, demonstrating CONOP feasibility and resulting in intensified operational use of system

Leadership

- Assistant Group Supervisor for Commercial Device Operations Group with 40+ staff members creating offensive software capabilities against endpoint devices; setting technical strategy, business development; driving staff development, salary, and performance reviews.
- Led $1.4M research TTO examining novel approaches for CNE. Responsible for setting technical direction, assigning tasking, managing staff and funding, and communicating results with sponsor
- Assistant Group Supervisor for Wireless Capability Group with 40+ staff members focused on RF collection and SDR; setting technical strategy, business development; driving staff development, salary, and performance reviews.
- Led five different multi-disciplinary teams including proof of concept and operational systems. Team sizes varied from 4-11 engineers with capabilities including SW development, RF, digital signal processing, and sys administration.
- Section-level Supervisor for 5-11 staff creating TTIL and specialized RF communications devices. College resume reviewer, primary interviewer, conducted performance reviews, counseled performance issues

- **Communications**
  - *Written:* Internally peer-reviewed deliverable research documents, SOWs, white papers, proposals, system test plans, test procedures, test reports, user manuals, performance evaluations, external awards submissions, and promotion packages.
  - *Oral:* Crafted and presented Group strategy for sector leadership, technical briefings to Laboratory management, capability briefings to flag-level leadership, dozens of technical status briefings to sponsor including future task direction, status, and capability delivery, tens of informal technical demonstrations and briefings for visiting sponsors

Lockheed Martin Integrated Systems and Solutions

Hanover, MD

Systems Engineer

May 2001 - Dec. 2006

- **Technical**
  - Developed embedded assembly language software for interoperability tests on telecommunications HW. Utilized reverse engineering skills to understand and modify undocumented systems. Performed full HW/SW life-cycle support. Reduced development time from 3 months to 6 weeks on a high profile assignment.
  - System engineer productizing customer prototypes. Captured system requirements in DOORS, documented transition steps and resources.
- **Advanced Technical Leadership Program (ATLP):** 2-year rotational program to develop high-potential technical leaders
  - *Rotation 1:* System engineer and TTO lead of 3-person team to productize customer prototypes to the APMEC contract.
  - *Rotation 2:* Business development - managing AFCEA symposium registration. Deputy Project Manager of corporate Value Proposition effort
  - *Rotation 3:* Project engineer managing schedule and funding for SOA task force.
- **Project Leadership**
  - Captured and conveyed customer staffing requirements to LM, created staffing requisitions, and interviewed candidates for new positions, resulting in 11 contractors added to the APMEC contract. Provided input to LM response for two new TTOs because of her teams efforts.
  - Tracked schedule and budget and conducted program status reviews for high visibility 30 FTE Market Research project

Honors & Awards

International Awards

2016 **Advocating Women in Engineering Award**, Society of Women In Engineering (SWE) Annual Conference *Philadelphia, PA*

Honors an individual who has demonstrated professional excellence in their chosen STEM field and has proven to be an advocate of women in engineering (5 awarded annually)

Presentation

WE and WE Local conferences

various

Society of Women In Engineering Conference Presenter

2018 - 2021

- Presented two years to job seekers regarding how to navigate the hiring process
- Moderated panel regarding improving chances in job applications
- Presented on work-life balance and maintaining job performance

XFIL CNO 2026 Conference

Baltimore, MD

Panelist

Sept. 2026

- Panelist discussing Non-kinetic effects and convergence

Program Committees

2018 - 2021 **Society of Women In Engineering Awards Judge**, WE and WE Local conferences *online*